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Revolutionizing Missing Data Handling with RFKFCM: Random Forest-based Kernelized Fuzzy C-Means

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Abstract

Missing values are a prevalent issue, frequently leading to a considerable decline in the quality of data. Therefore, it becomes imperative to adeptly manage missing data. This study presents a technique for imputing missing values that incorporates Kernelized Fuzzy C-Means (KFCM) clustering. The technique introduced is termed RFKFCM, which combines the benefits of Random Forest (RF) and KFCM algorithms. The proposed RFKFCM's performance is evaluated through a comparative analysis with five state-of-the-art imputation techniques (mean, LI, KNN, FCM, and LIFCM) across four widely used real-world datasets obtained from the UCI repository. Additionally, experiments investigating the impact of missing values affirm the robustness of the proposed technique across different missing rates using Friedman's statistical test for each dataset by comparing their mean rank. The experimental results indicate that our proposed technique performs significantly better than the existing imputation techniques based on Root mean squared error (RMSE) & Mean absolute error (MAE) for these datasets.

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Keywords: Missing values; RF; KFCM; Missing data; Fuzzy clustering; MVI; RFKFCM.

1. Introduction

The existence of missing values within datasets has consistently posed a significant challenge in the field of machine learning. It may occur in a database due to variety of causes: tool defects, unresponsiveness, data entry errors,

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undetermined value, tools scarcity, data inconsistency, shortage of time, etc. [1]. Consequently, the decision-making process is hindered, as the absence of holistic grasp of the data renders it less effective and impactful.

The mechanisms that lead to missing data can be categorized into three distinct groups: (i) Missing Completely at Random (MCAR); (ii) Missing at Random (MAR); and (iii) Not Missing at Random (NMAR) [2]. MCAR occurs when the likelihood of a data point having a missing value for a specific feature is unrelated to both the known and missing values in the dataset. In contrast, if the likelihood of a data point having a missing value for a particular feature is influenced by the observed value (known values), it is termed MAR. Moreover, when it comes to NMAR, missing data is contingent upon the actual value of the feature [3].

Researchers have proposed a diverse array of techniques for tackling the issue of missing values which ranges from eliminating instances with missing values to employing different imputation techniques. Even though the deletion technique is simple in the way that instances with missing cases are deleted, it creates a problem of reduction in sample size and may lead to providing an invalid conclusion [2, 4].

To achieve improved outcomes, a more efficient approach involves replacing missing values with predicted values rather than removing them. Missing Value Imputation (MVI) techniques are employed for this purpose, which can be broadly categorized into two groups: single imputation (SI) and multiple imputation (MI) [4]. In SI, a single substituted value is predicted for imputation [4]. However, these techniques prove to be ineffective and cause standard errors and biased results as they do not consider the randomness of incomplete data. Some of the examples of SI techniques include mean/mode, hot deck, k-nearest neighbor (KNN), and linear interpolation (LI) imputation techniques. Researchers have experimentally shown that the SI techniques cannot accurately determine the actual values [4].

In the context of multiple imputation (MI), multiple predicted values can be employed for imputation [4]. These techniques include uncertainty and provide unbiased results [4]. MI encompasses a variety of techniques, including decision tree, random forest (RF), k-means clustering, and Fuzzy clustering. Out of the various MVI techniques, the most efficient one is the fuzzy clustering imputation technique. Different variations of fuzzy clustering techniques have been utilized by several researchers for dealing with missing data such as Fuzzy c-means (FCM) [1, 5, 6, 16] and Kernelized FCM(KFCM) [7]. This study focuses on utilizing RF and a fuzzy clustering technique for handling missing values.

Random forest utilizes a combination of decision trees for the prediction of missing values [8]. KFCM exhibits a superior ability to handle noise and outliers in comparison to FCM and Possibilistic Fuzzy c-means (PFCM) fuzzy clustering techniques [7]. This paper proposed an imputation technique, Random Forest-based Kernelized Fuzzy c-means (RFKFCM), which combines the advantages of RF (ensemble imputation technique) and KFCM (MI technique) for the imputation of missing values.

The primary contributions of this study are threefold: firstly, the proposal of a new hybrid imputation technique named RFKFCM. Secondly, empirical validation underscores that RFKFCM surpasses existing state-of-the-art MVI techniques in terms of imputation performance. Thirdly, the application of Friedman's statistical test for ranking MVI techniques reveals that our proposed approach consistently secures the highest rank, thus affirming its robustness across various missing data rates.

The rest of this paper is organized as follows. Section 2 briefly presents the related work. Section 3 describes the proposed RFKFCM imputation technique. Section 4 presents the experimental results & discussion of comparing RFKFCM with other imputation techniques. Section 5 provides a conclusion for the paper and discusses future work.

2. Related Work

Researchers have undertaken various surveys to investigate the impact of MVI approaches on performance measures [3, 9, 10].

Schmitt et al. [11] showed a comparison of the six most popular MVI techniques for missing data imputation namely Mean, KNN, fuzzy K-means (FKM), singular value decomposition (SVD), Bayesian principal component analysis (bPCA), and multiple imputations by chained equations (MICE). This comparison was done using four real datasets. Different performance measures like root mean square error (RMSE), un-supervised clustering error, and

supervised clustering error were used for calculating the effectiveness of these techniques. Their study concluded that the bPCA and FKM techniques outperformed all the other techniques.

In a study conducted by Jadhav et al. [12], seven imputation techniques were applied to numeric data to conduct a comparative analysis. These techniques encompass mean imputation, predictive mean matching, Bayesian linear regression (norm), median imputation, non-Bayesian linear regression (norm. nob), kNN, and random sampling. The study drew from five numeric datasets sourced from the UCI machine learning repository to facilitate the comparative assessment. The results of the study led to the conclusion that among all the techniques evaluated, kNN imputation exhibited the highest performance.

Patil et al. [13] analyzed the performance of two major classification algorithms namely Naive Bayes and J48 for data classification for missing value prediction using a bank dataset. They considered accuracy, specificity, and sensitivity for comparison, along with utilizing correct and incorrect examples that provide the most effective technique for classification. Authors in [14] presented a survey that reviewed and compared some MVI approaches. This paper illustrated the influence of the missing rate and data type on the performance and efficiency of the imputation technique.

Hathway and Bezdek [5], in their pioneering work in 2001, introduced four distinctive approaches for addressing missing data using the Fuzzy C-Means (FCM) technique. These novel strategies included the Whole Data Strategy (WDS), Partial Data Strategy (PDS), Optimal Completion Strategy (OCS), and Nearest Prototype Strategy (NPS), marking significant contributions to the field of handling incomplete data.

In [7], a kernel technique based on a classical fuzzy clustering algorithm was proposed for clustering incomplete data. They replaced the Euclidean distance matric with the kernel-induced matric in the data space. Three different kernel functions namely Gaussian, RBF function, and hyper tangent function were applied. In their studies, several researchers have identified attribute types [15], and correlation [16] as factors for handling missing values. Hu et al. proposed a technique that utilizes similarity measures to mitigate the impact of outliers during the processing of missing data [17].

An alternative approach based on FCM, which employs a data-filling approach for handling incomplete soft sets, was introduced as ADFIS [18]. This approach was designed for effectively managing incomplete soft sets. The study encompassed experiments on four distinct datasets, and the findings ultimately concluded that ADFIS consistently outperforms all other techniques in terms of accuracy.

In the domain of medical applications, Kumaran et al. introduced an innovative approach that combined hybrid Fuzzy C-Means (FCM) with majority vote techniques to heighten the precision of predicting missing values in microarray data [10]. Their research particularly emphasized the critical importance of accuracy in this field. To gauge the effectiveness of their method, they conducted performance comparisons against three other imputation techniques: zero imputation, k-Nearest Neighbors (kNN), and standard FCM. The results led them to assert the superiority of their proposed approach. In a related context, Purandhar et al. introduced an enhanced FCM approach tailored for applications within the medical domain, further advancing the state of the art in this field [19].

3. Proposed Technique

This paper proposed a novel technique, Random Forest-based Kernelized Fuzzy c-means (RFKFCM) for MVI which combines the strengths of KFCM and RF techniques. Fig. 1 outlines the flowchart detailing the proposed technique. The steps of proposed RFKFCM technique are described as follows:

Step 1. Generating Incomplete data

From an entire dataset $\boldsymbol{D}$, an incomplete dataset $\boldsymbol{D_w}$ is formed by randomly removing a specific percentage of values.

Step 2. Utilize Random Forest imputation to fill in missing values

Random Forest (RF) can be used for identifying missing data with better similarity and higher correlations, as a random forest can deal with both categorical and continuous data. By averaging multiple deep decision trees, RF can generally improve the performance in the final model, with a reduction of the variance decision trees to make a more accurate and stable prediction than individual trees [12].

Step 3. Employ KFCM on the entire dataset:

The KFCM technique groups data-items from a dataset X = $\{x_1, x_2, \ldots . x_n\}$ into fuzzy clusters $v = \{v_1, v_2, \ldots . v_c\}$ as specified in Eq. (1) by the minimization of the following objective function [11]:

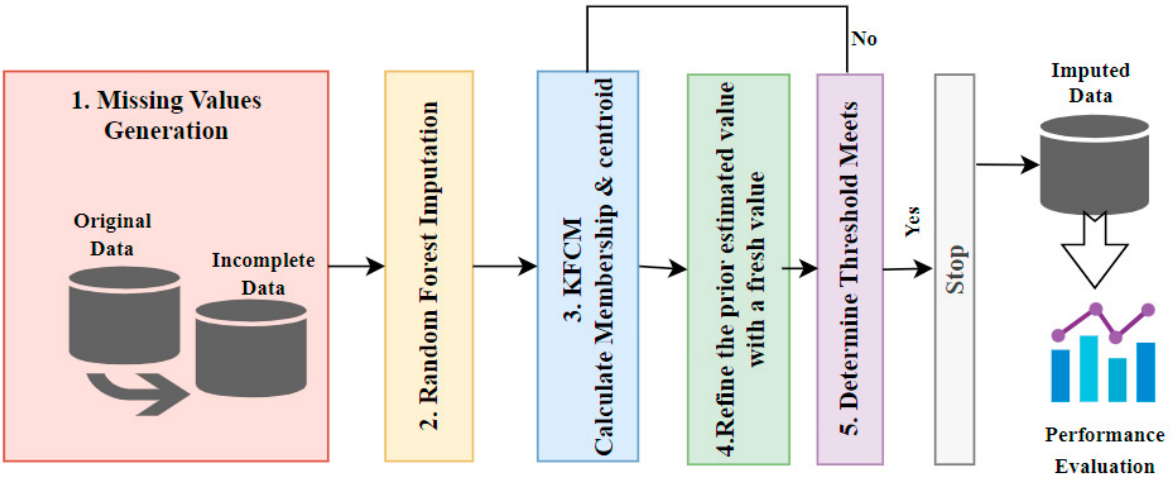


Fig. 1. Flowchart of the proposed RFKFCM technique

$$\boldsymbol{J_{FCM}} = \sum_{j=1}^{c}\sum_{p=1}^{n} u_{jp}^{m}||\phi(\boldsymbol{x}_p) - \phi(\boldsymbol{v}_j)||^2 \qquad (1)$$

Where 'n', 'm', and 'c' denote the number of data-items, fuzzifier (constant value), and number of clusters respectively. And 'u_{ik}' signifies fuzzy memberships of data-item 'x_k' in cluster i as given in Eq. (2).

$$\sum_{\boldsymbol{j=1}}^{\boldsymbol{c}} \boldsymbol{u}_{jp} = 1, \;\; \boldsymbol{u_{jp}} \in [0,1], for \;\; p = 1,2,3,\cdots,n \qquad (2)$$

$\phi_{d_{pj}^2} = ||\phi(\boldsymbol{x}_p) - \phi(\boldsymbol{v}_j)||^2$ is the square of distance which denote the distance between $\phi(\boldsymbol{x}_p)$ and $\phi(\boldsymbol{v}_j)$. Each data item's membership is adjusted using the following equation:

$$u_{jp} = \frac{1}{\sum_{j=1}^{c}\left(\frac{\phi_{d_{pc}^2}}{\phi_{d_{pj}^2}}\right)^{\frac{1}{(m-1)}}} \qquad \forall \boldsymbol{j,p} \qquad (3)$$

The centroids are updated with Eq. (4) as follows:

$$v_j = \frac{\sum_{p=1}^{n} u_{jp}^{m} x_p}{\sum_{p=1}^{n} u_{jp}^{m}} \qquad (4)$$

Step 4. Refine the prior estimated value with a fresh one

In this phase, the earlier approximated missing values undergo an update with fresh values. Let 'x_{sa}' represent the ath attribute value of sth instance within dataset X. If 'x_{sa}' denotes a missing value, its calculation can be carried out using Eq. (5) as outlined below:

$$x_{sa} = \sum_{j=1}^{c} u_{ja}^{m} v_{sj} \qquad (5)$$

Step 5: Determining the Threshold

Repeat step 3 and 4 until threshold meets. Threshold is specified as: $\|v^{(i+1)} - v^{(i)}\| < \epsilon$, Where ϵ =0.00001 [20].

4. Experimental Result and Discussion

This section describes the datasets and performance metrics utilized in this study. Furthermore, experimental results have been discussed that were performed on MATLAB R2022a software. Additionally, obtained results have been validated on SPSS tool.

4.1. Dataset Description

The proposed RFKFCM's performance is evaluated across four extensively utilized datasets, which are sourced from the UCI repository [21]. For a more comprehensive understanding of these datasets, the descriptive details such as no. of instances, attributes, and classes have been furnished in Table 1.

Table 1. Description of datasets.

Name	No. of instances	No. of attributes	No. of classes
Iris	150	4	3
Glass	214	9	2
Seed	210	7	3
Wine	178	13	3

For experimentation, a random selection of data from the dataset has been deliberately removed and designated as missing values. These missing values are estimated through a variety of imputation techniques, encompassing mean [12], LI [22], KNN [12], FCM [5], LIFCM [23], and our proposed RFKFCM across different missing rates of 1%, 3%, 5%, 7%, and 9%.

4.2. Performance Metrics

The performance of RFKFCM is evaluated by employing widely recognized performance metrics: Root mean squared error (RMSE) and Mean absolute error (MAE) [24].

$$\text{RMSE}= \sqrt{\frac{1}{m}\sum_{p=1}^{m}[E_p - A_p]^2} \qquad (6)$$

$$\text{MAE}= \frac{1}{m}\sum_{p=1}^{m}|E_p - A_p| \qquad (7)$$

Here, m is the number of missing values, E_p and A_p ($1 \le p \le m$) are estimated and actual values, respectively, for the pth missing value. Values of RMSE and MAE range between 0 to ∞, with a lower value denoting enhanced imputation performance.

4.3. Experimental Results for imputation technique

We compare RFKFCM with four state-of-art MVI techniques namely mean, LI, KNN, FCM, and LIFCM across five different missing rates. In total, there are 20 combinations of missing data, resulting from the combination of four datasets at five different missing rates.

The quantitative assessment of mean, LI, KNN, FCM, LIFCM, and RFKFCM's performance across all datasets is presented in Tables 2 and 3. These tables provide insights based on RMSE and MAE metrics and encompass 20 different missing combinations. This analysis comprehensively evaluates each technique's effectiveness in handling missing data across various scenarios. Furthermore, it's worth noting that among the six techniques, the best results

have been highlighted for clarity. For instance, in Table 2, when considering a 1% missing rate for the Seed dataset, RFKFCM achieves an impressive RMSE of 0.058. In comparison, the RMSE values for mean, LI, KNN, FCM, and LIFCM with the same 3% missing rate for the Seed dataset are 0.128, 0.122, 0.111, 0.086, and 0.088 respectively. Observations indicate that RFKFCM consistently outperforms other techniques across a majority of the missing combinations based on RMSE and MAE.

The graphical illustration of RMSE, and MAE for Iris, Glass, Seed, and Wine dataset is shown in Figs. 2 and 3. It is evident from the data presented in Tables 2 and 3, as well as the visual insights from Figures 2 and 3, that the RFKFCM missing value imputation technique consistently yields superior results in comparison to the mean, LI, KNN, FCM, and LIFCM imputation techniques. These results were observed across the four widely-used standard datasets. The experimental findings lead to the conclusion that the proposed technique exhibits greater efficiency when applied to real-world datasets across varying missing rates.

Table 2. Comparison of performance of the proposed technique with other MVI techniques on datasets in terms of RMSE.

Dataset	MVI Technique	Missing Rate[†1]				
		1%	3%	5%	7%	9%
Iris	**Mean**	0.376	0.382	0.674	0.499	0.735
	LI	0.283	0.379	0.544	0.488	0.656
	KNN	0.321	0.312	0.621	0.682	0.740
	FCM	0.284	0.326	0.574	0.659	0.573
	LIFCM	0.168	0.425	0.485	0.580	0.588
	RFKFCM	**0.125**	**0.240**	**0.403**	**0.379**	**0.407**
Glass	**Mean**	0.084	0.128	0.171	0.198	0.199
	LI	0.083	0.094	0.148	0.171	0.173
	KNN	0.052	0.075	0.125	0.187	0.196
	FCM	0.099	0.082	0.113	0.135	0.144
	LIFCM	0.053	0.074	0.118	0.159	0.126
	RFKFCM	**0.029**	**0.065**	**0.098**	**0.120**	**0.124**
Seed	**Mean**	0.128	0.109	0.143	0.163	0.195
	LI	0.122	0.128	0.115	0.110	0.134
	KNN	0.111	**0.097**	0.131	0.142	0.164
	FCM	0.086	0.166	0.124	0.125	0.154
	LIFCM	0.088	0.144	0.121	0.143	0.122
	RFKFCM	**0.058**	0.098	**0.081**	**0.109**	**0.117**
Wine	**Mean**	0.188	0.240	0.244	0.2	0.244
	LI	0.193	0.197	0.140	0.187	0.184
	KNN	0.176	0.221	0.104	0.121	0.137
	FCM	0.154	0.210	0.102	0.121	0.150
	LIFCM	0.154	0.233	0.132	0.188	0.199
	RFKFCM	**0.143**	**0.186**	**0.098**	**0.109**	**0.119**

[1] The best result among the six imputation methods is denoted by bold values in the table.

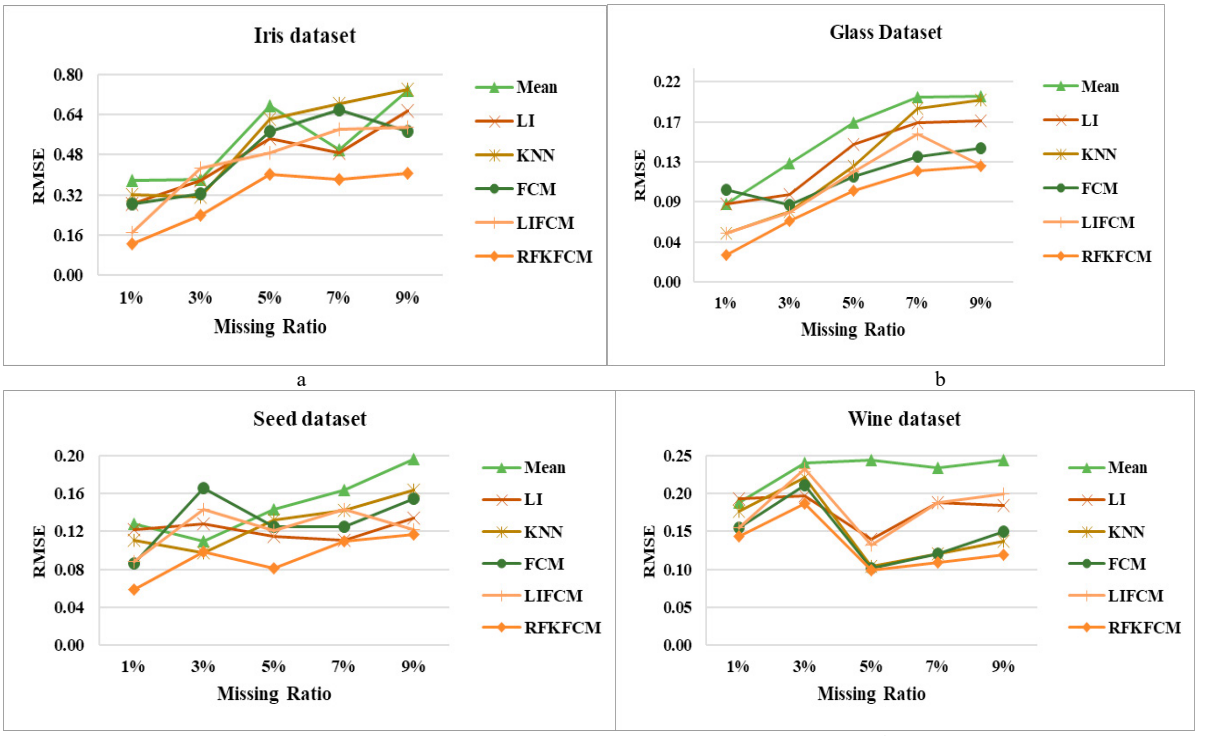


Fig. 2. Comparison of performance analysis of proposed imputation technique with other MVI techniques on (a) Iris dataset, (b) Glass dataset, (c) Seed dataset, and (d) Wine dataset in terms of RMSE (the lower the better)

Table 3. Comparison of performance of proposed technique with other MVI techniques on datasets in terms of MAE.

Dataset	MVI Technique	Missing Rate				
		1%	3%	5%	7%	9%
Iris	**Mean**	0.252	0.234	0.545	0.465	0.668
	LI	0.154	0.242	0.377	0.439	0.645
	KNN	0.192	0.256	0.452	0.429	0.547
	FCM	0.154	0.177	0.367	0.372	0.484
	LIFCM	0.140	0.168	0.323	0.380	0.396
	RFKFCM	**0.119**	**0.131**	**0.298**	**0.282**	**0.312**
Glass	**Mean**	0.039	0.052	0.096	0.110	0.121
	LI	0.042	0.044	0.080	0.085	0.089
	KNN	0.031	0.025	0.055	0.091	0.113
	FCM	0.042	0.027	0.046	0.060	0.089
	LIFCM	0.017	0.020	0.030	0.043	0.052
	RFKFCM	**0.009**	**0.019**	**0.027**	**0.031**	**0.027**
Seed	**Mean**	0.099	0.087	0.099	0.112	0.143
	LI	0.075	0.077	0.091	0.089	0.110
	KNN	0.055	0.066	0.088	0.109	0.137
	FCM	0.034	0.125	0.081	0.096	0.115
	LIFCM	0.055	0.116	0.098	0.110	0.127
	RFKFCM	**0.032**	**0.063**	**0.057**	**0.087**	**0.098**
Wine	**Mean**	0.155	0.177	0.188	0.184	0.202
	LI	0.144	0.177	0.133	0.143	0.155
	KNN	0.133	0.166	0.098	0.088	0.098
	FCM	0.099	0.166	**0.087**	0.077	0.098
	LIFCM	0.100	0.199	0.110	0.132	0.144

	RFKFCM	**0.095**	**0.138**	0.088	**0.066**	**0.075**

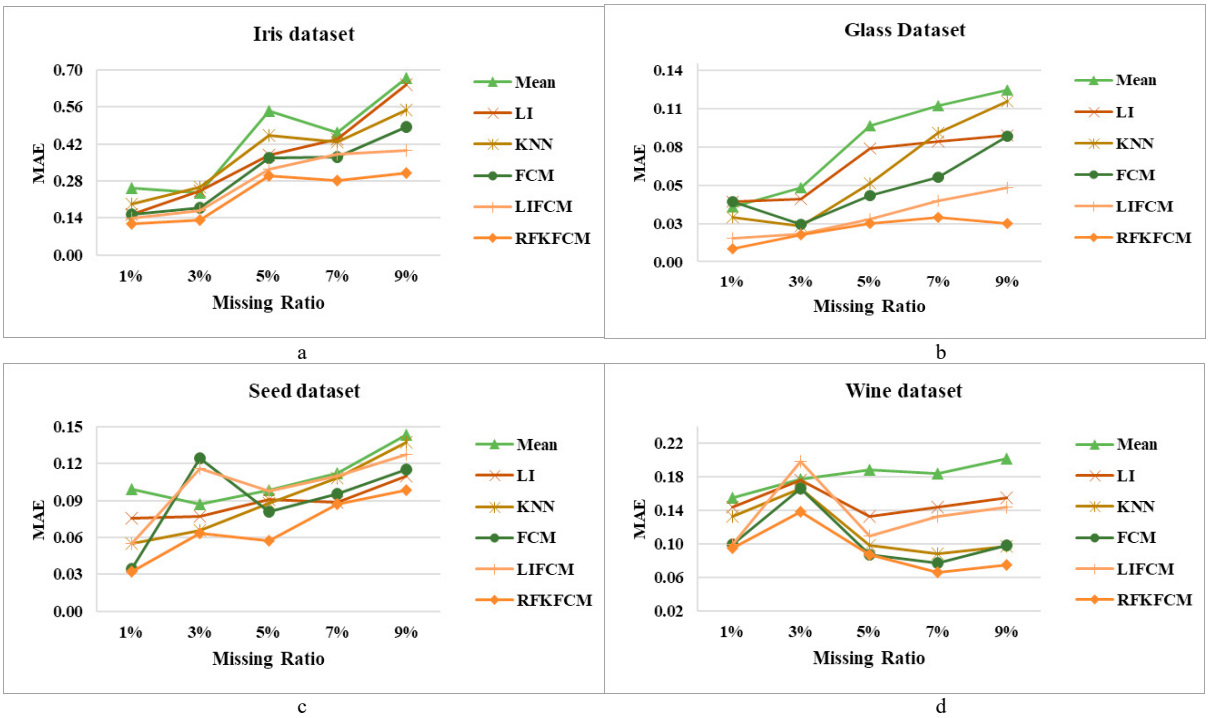


Fig. 3. Comparison of performance analysis of proposed imputation technique with other MVI techniques on (a) Iris dataset, (b) Glass dataset, (c) Seed dataset, and (d) Wine dataset in terms of MAE (the lower the better)

4.4. Friedman's statistical test for comparing multiple MVI techniques

Friedman test is a non-parametric statistical method used for the comparison of multiple results [25]. In this study, this test has been utilized for assessing the performance of MVI techniques across different datasets based on RMSE, MAE. The test assigns rank to each imputer, thereby quantifying how well each imputation technique has performed. The results are depicted in Tables 4 and 6, showcasing the different Friedman's statistics. In these Table, N defines the total number of missing combinations considered.

Table 4. Friedman's test statistics for comparing RFKFCM and state-of-art imputation techniques across different datasets for RMSE

Statistics	**Iris**	**Glass**	**Seed**	**Wine**	**Overall**
N	5	5	5	5	20
Chi-square	14.714	18.943	13.114	19.538	58.617
Degree of freedom	5	5	5	5	5
P- value	0.012	0.002	0.022	0.002	2.35E-11

The outcomes of the Friedman tests, as reflected by the assigned ranks, are presented in both Table 5 and 7.

Table 5. Ranks obtained through Friedman's testing across different datasets for RMSE

MVI Technique	Iris	Glass	Seed	Wine	Final Mean Rank
Mean	5	5.8	5.4	5.8	5.5
LI	3.2	4.4	3.2	4.2	3.75
KNN	4.8	3.8	3.8	2.9	3.88
FCM	3.6	3.4	3.8	2.8	3.35
LIFCM	3.4	2.6	3.6	4.3	3.48
RFKFCM	1	1	1.2	1	1.05

Table 6. Friedman's test statistics for comparing RFKFCM and state-of-art imputation techniques across different datasets for MAE

Statistics	Iris	Glass	Seed	Wine	Overall
N	5	5	5	5	20
Chi-square	22.126	20.896	17.011	21.919	67.042
Degree of freedom	5	5	5	5	5
P-value	0.000495	0.000848	0.004478	0.000543	4.22E-13

Table 7. Ranks obtained through Friedman's testing across different datasets for MAE

MVI Technique	Iris	Glass	Seed	Wine	Final Mean Rank
Mean	5.6	5.6	5.6	5.7	5.63
LI	4.5	4.6	3.2	4.9	4.3
KNN	4.8	4	3.5	3	3.83
FCM	2.9	3.8	3.2	2	2.98
LIFCM	2.2	2	4.5	4.2	3.23
RFKFCM	1	1	1	1.2	1.05

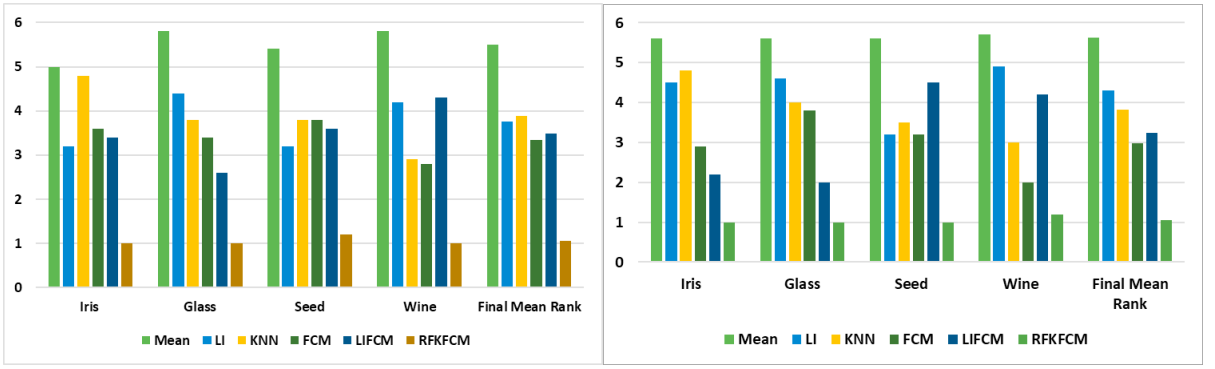


Fig 4. Comparison of RFKFCM and state-of-art imputation techniques on the basis of ranks obtained through Friedman's testing across different datasets for a) RMSE and b) MAE

The findings of the Friedman's test, as depicted in Figure 4, unequivocally highlight RFKFCM as the top-performing Missing Value Imputation (MVI) technique. It consistently secures the highest rank across all datasets, excelling in both RMSE and MAE criteria. In stark contrast, the mean imputation method ranks lowest, indicating its inferior performance.

5. Conclusion & Future work

Missing values have been encountered as major obstacle in research studies, hindering valid conclusions. Addressing this, various research works are available in the field of MVI (single as well as multiple imputations) aim to enhance data quality.

This paper introduces a technique for handling missing data through the integration of KFCM and RF techniques named RFKFCM. The proposed approach has been compared against five other existing MVI techniques across four publicly accessible real-world datasets, with a focus on two performance metrics: RMSE and MAE.

According to the experimental results, the RFKFCM technique outperforms the other five state-of-the-art techniques by a significant margin in both RMSE and MAE metrics. Furthermore, Friedman's statistical tests were performed on the MVI techniques to find out the best imputation technique across all datasets. The results showed that the proposed RFKFCM performed relatively much better than other imputation techniques in terms of RMSE and MAE by achieving a rank of 1.05. Hence, the proposed technique demonstrates its efficacy in effectively imputing missing values across all the datasets with different missing rates. In future studies, there is a potential for conducting further research to investigate its suitability across a wider range of datasets and applications, in order to assess its potential for generalization.

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